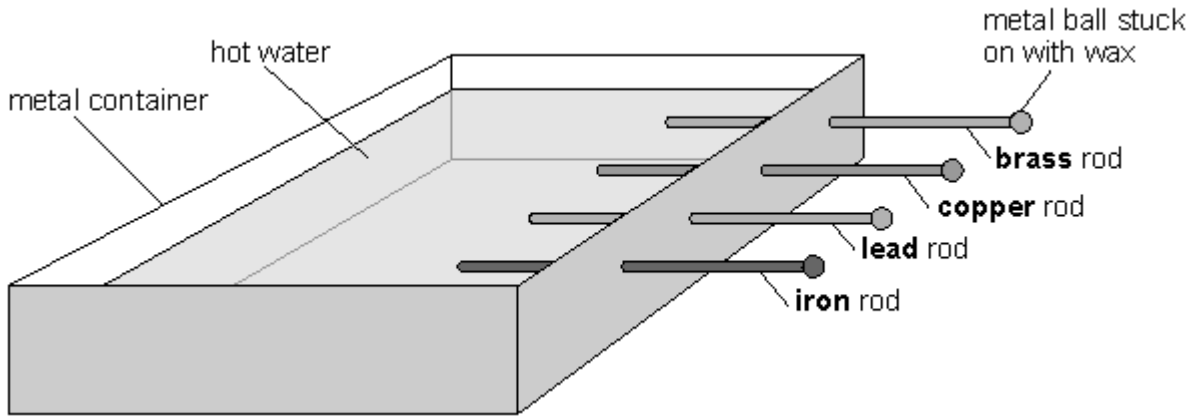
Q1.

Leanne had four rods, each made from a different metal.
She wanted to find out which metal was the best conductor of heat.
The diagram shows some of Leanne's equipment.



(a) Leanne's results are shown in the table.

metal rod	time for metal ball to drop off (seconds)
brass	36
copper	24
lead	246
iron	132

What measuring equipment did Leanne use to get her results?

.....

1 mark

(b) Give **two** things Leanne must do to carry out a fair test.

1 .....

2 .....

2 marks

(c) Which metal in the table was the best conductor of heat?
Tick the correct box.

brass	☐	copper	☐
iron	☐	lead	☐

1 mark

(d) Leanne left the rods in the water for a week.
One of the metal rods went rusty.

Which metal rod went rusty?
Tick the correct box.

brass	☐	copper	☐
iron	☐	lead	☐

1 mark
maximum 5 marks

Q1.

(a) • stopwatch

accept 'stopclock'
accept 'timer'
'timer' is accepted here and not in Paper 1 question 1, as 'time' is not mentioned in this question
accept 'clock with a second hand'
'clock' is insufficient

1 (L3)

(b) any **two** from

• same length of rods

accept for one mark 'the rods are the same
*'same temperature **or** volume **or** container **or** rod' is insufficient*

• same thickness of rods

*size' if not given with either 'same length' **or** 'same thickness'*

• same sized balls **or** same metal balls

• same amount of wax

• same type of wax

• the ends of all rods (in the water) are at the same temperature

accept 'same length (of rod) in water'

• she put the water in after sticking the balls on

2 (L4)

(c) • copper ✓

if more than one box is ticked, award no mark

1 (L4)

(d) • iron ✓

if more than one box is ticked, award no mark

1 (L4)

[5]